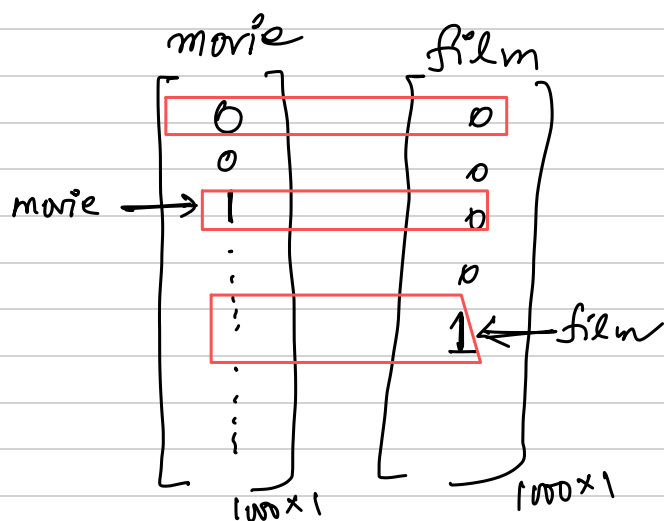


The problem with one-hot vectors

- ① if the vocabulary is very long, each token will have a one-hot vector that's as long as the size of the vocabulary
- ② Consider the one-hot-encoded vectors for "movie" & "film"
⇒ Are these two vectors "close" to each other?



Closeness = Distance
= Euclidean Distance
 $\sqrt{1+1} = \sqrt{2}$

The distance between any two one-hot vectors $\sqrt{2}$

Word Embedding

- Stand-alone word embeddings
- Contextual word embeddings via Transformers!

Stand-alone word embeddings

- manually collect synonyms, antonyms, related words etc and try to assign embedding vectors to them that satisfy the requirements.

The acting in the _____ was superb

movie ✓

film ✓

musical ✓

⋮

truck ✗

banana ✗

tensor ✗

if {movie, film and musical} appears in the same contexts (i.e. sentences) very often, they are likely to be related.

⇒ More generally, related words appear in similar context

Imag: look all sentences from Wikipedia

Glove Vectors
Word 2 vec

The GloVe Model

- We assume that the co-occurrence count of a word pair is a (simple) linear function of the two biases and the two embedding vector as follows:

$$X_{ij} = b_i + b_j + W_i^T W_j$$

$$\log(X_{ij}) = b_i + b_j + W_i^T W_j$$

How can we learn the weights of this model?

$$\text{Minimize } \sum_{i,j} [\log(X_{ij}) - (b_i + b_j + W_i^T W_j)]^2$$